Multi-Source Candidate Data Transformer

One-page technical design: extraction, normalization, entity resolution, merge, confidence, projection, validation

Extract – Normalize – Resolve – Merge – Score – Project – Validate

Goal

Fuse CSV, ATS JSON, resume PDF or TXT, and recruiter notes into one clean canonical candidate profile. The pipeline is deterministic and explainable: every output field has a selected value, confidence score, and source provenance when enabled.

Canonical Schema

full_name, email, phone, country, skills, date_of_birth, and experience_yrs. Emails are lowercased, phones are E.164, countries are ISO alpha-2, dates are YYYY-MM-DD, names are cleaned/title-cased, and skills use a synonym map before dedupe and sort.

Entity Resolution

Records first match on exact normalized email, then exact phone, then blocked fuzzy name matching. Name matching accepts exact normalized names, same first name with matching last initial, or high similarity. Conflicting countries block weak fuzzy merges.

Merge Rules

Names prefer the longest normalized value. Emails/countries use support count then source priority. Phones use support, normalized digit length, then priority. Skills are unioned. Experience uses the maximum reported years. Source priority is ATS, CSV, Resume, Notes.

Projection and Validation

Runtime JSON config selects fields, renames fields, toggles confidence/provenance, and controls missing data as null, omit, or error. The projected JSON is validated with a dynamic Pydantic model before output.

Confidence Heuristic

Evidence	Scalar Fields	Skills
3+ supporting sources	0.95	0.85
2 supporting sources	0.90	0.75
1 supporting source	0.62 + priority factor	0.60